ASSIGNMENT–02: COMPARATIVE STUDY AND PRACTICAL ANALYSIS OF GENERATIVE AI MODELS

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1. Objective and Experimental Setup

Objective: To study and analyse image reconstruction using Autoencoder (AE) and Variational Autoencoder (VAE) (CO3), and image generation using Deep Convolutional GAN (DCGAN) and a pretrained Diffusion Model, Stable Diffusion (CO4).

The executed notebook uses Fashion-MNIST (28 × 28 grayscale clothing images). AE and VAE are evaluated on the same test images. AE is trained for 10 epochs; VAE for 12 epochs; DCGAN for 15 epochs. Stable Diffusion v1.5 is loaded with pretrained weights and generates a 512 × 512 image from a text prompt using 20 inference steps and guidance scale 7.0. The notebook reports no GPU devices and CUDA unavailable for PyTorch; Stable Diffusion therefore ran on CPU.

2. Comparative Study of Four Models

Aspect	Autoencoder (AE)	Variational AE (VAE)	DCGAN	Stable Diffusion
Purpose	Reconstruct input	Reconstruct; learn probabilistic latent space	Generate new images	Generate images from text
Working / components	Encoder → latent code → decoder	Encoder predicts μ and log-variance; sampling + decoder	Generator creates samples; discriminator classifies real/fake	Text encoder, denoising U-Net, scheduler and VAE decoder
Advantages	Simple; useful for compression and denoising	Structured latent space; sampling and interpolation	Convolutional image synthesis	Prompt-guided, detailed, high-resolution output
Limitations	May lose fine details	Can produce smoother / blurrier reconstructions	Training instability and artifacts	Compute-intensive; prompt adherence may vary
Applications	Denoising,	Representation	Synthetic image	Text-to-image

	compression, anomaly detection	learning and sampling	generation	design and content creation
Output in this experiment	Reconstruction of test images	Reconstruction using posterior mean	16 new 28×28 grayscale samples	One prompt-based 512×512 RGB sneaker image

Dataset example images are shown below; the same Fashion-MNIST dataset was used for AE, VAE and DCGAN.

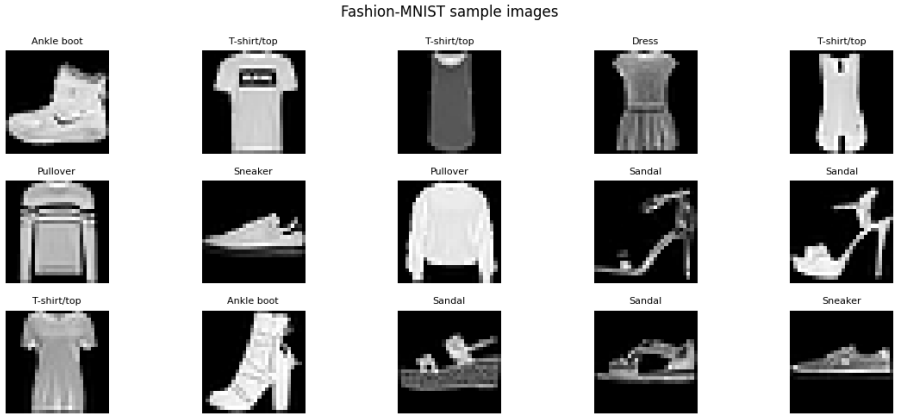


Figure 1. Fashion-MNIST sample images displayed by the executed notebook.

3. Practical Activity A – Image Reconstruction (CO3)

The notebook compares original test images with AE and VAE reconstructions. AE uses a deterministic latent code. VAE predicts a latent mean and log-variance, samples through the reparameterization trick during training, and combines reconstruction loss with KL-divergence. For the displayed VAE reconstructions, the posterior mean is used.

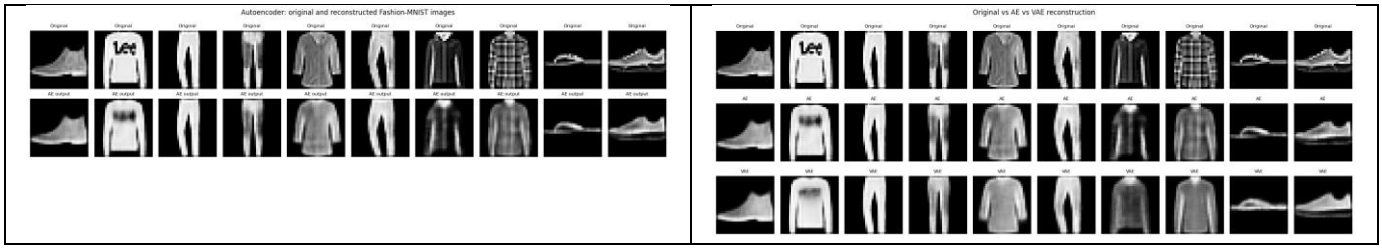


Figure 2. AE reconstruction outputs (left) and original–AE–VAE comparison (right).

Quantitative results over the complete test split:

Model	Test MSE	Test MAE
Convolutional Autoencoder	0.006448	0.039754
Variational Autoencoder	0.009283	0.047689

Analysis: AE has lower MSE and MAE in this run, indicating smaller pixel-wise reconstruction error. Both models preserve the broad clothing shapes, while fine boundaries and details can appear softened. Pixel metrics do not fully measure perceived visual quality.

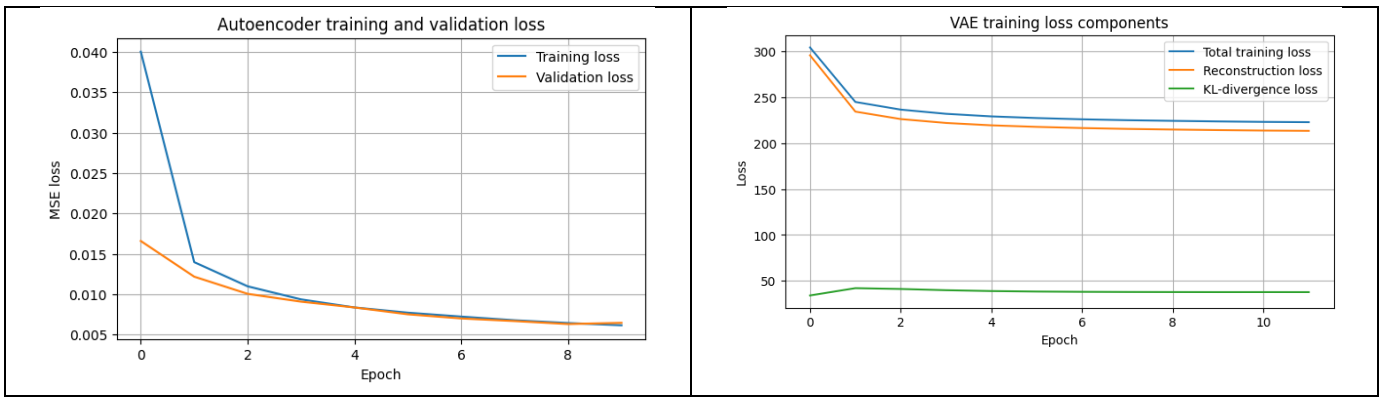


Figure 3. Notebook training plots: AE training/validation MSE loss (left); VAE total, reconstruction and KL losses (right).

The AE plot shows training and validation loss decreasing across 10 epochs. The VAE plot shows its total loss components over 12 epochs; the KL term regularizes the latent distribution.

4. Practical Activity B – Image Generation (CO4)

DCGAN: The generator maps random noise to 28 × 28 grayscale images, while the discriminator learns to distinguish real Fashion-MNIST images from generated ones. The notebook trains for 15 epochs and generates 16 samples using a fixed noise batch.

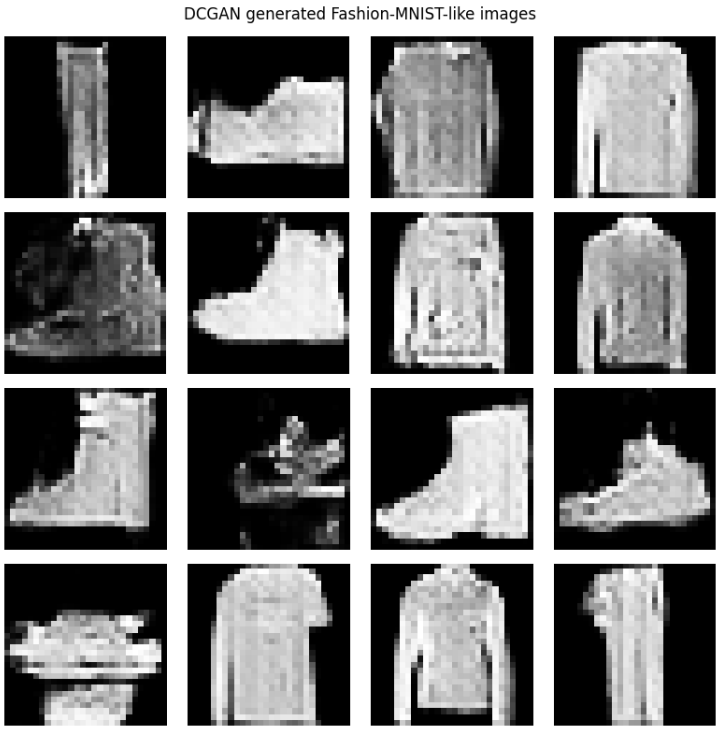


Figure 4. Actual DCGAN generated Fashion-MNIST-like sample grid (16 outputs).

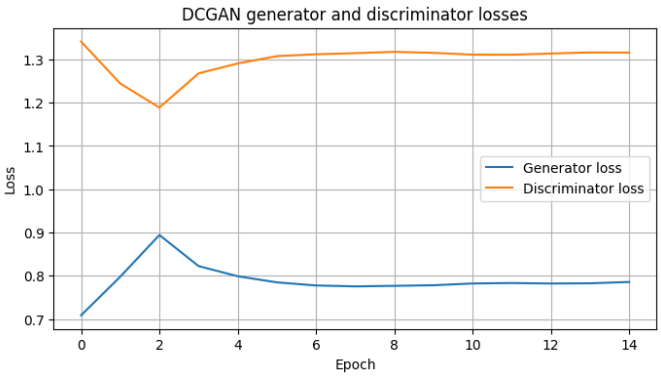


Figure 5. Actual DCGAN generator and discriminator loss curves over 15 epochs.

Analysis: The outputs contain clothing-like forms but vary in clarity and completeness. The loss curves show adversarial training behaviour; losses alone do not prove visual quality.

Stable Diffusion: The notebook loads the pretrained stable-diffusion-v1-5/stable-diffusion-v1-5 pipeline. Prompt: "A high-quality studio product photograph of a single classic white sneaker, side view, isolated on a clean light grey background, soft studio lighting, sharp details, realistic material texture." It uses a negative prompt to discourage blur, distortion, multiple shoes, cropping, watermark, text and logo.



Figure 6. Actual Stable Diffusion v1.5 output (512 × 512 RGB).

Analysis: Stable Diffusion follows the sneaker prompt and produces a detailed product-style image. It is not a Fashion-MNIST reconstruction or a model trained on Fashion-MNIST. Its output is compared qualitatively with DCGAN because the input conditions and resolutions differ.

5. AI Usage, Reflection and Contributions

Tools:

ChatGPT (explanations, code assistance and debugging); Google Colab (execution); TensorFlow/Keras (AE, VAE, DCGAN); PyTorch and Hugging Face Diffusers (Stable Diffusion). Prompts included requests to explain and compare the four architectures, implement Fashion-MNIST reconstruction, implement DCGAN generation, and run pretrained Stable Diffusion text-to-image inference.

Verification: checked dataset shape and pixel ranges; used the same test images for AE and VAE; calculated MSE/MAE on the test set; inspected reconstruction images and training plots; reviewed DCGAN samples and loss curves; and checked the Stable Diffusion result against its recorded prompt.

Learning outcomes: distinguished reconstruction from generation; understood deterministic and probabilistic latent representations, KL regularization, adversarial generator/discriminator learning, and prompt-conditioned diffusion.

Member contributions:

Sakshi – Studied AE/VAE, analysed reconstruction outputs and compared MSE/MAE, analysed generated images/prompts, and documented results.

Shreya – Worked on Fashion-MNIST preprocessing and DCGAN, analysed generated images/training results, and prepared the report and PPT.

Dhanashri – Studied and implemented Stable Diffusion, analysed generated images/prompts, and documented results.

6. Conclusion

The notebook demonstrates AE/VAE reconstruction and DCGAN/Stable Diffusion generation. AE obtained lower pixel-wise test errors than VAE in this experiment. DCGAN produced 28 × 28 grayscale amples; Stable Diffusion produced a prompt-conditioned 512 × 512 image. Architecture, training method, latent representation, prompt conditioning and compute resources affect the outputs.

References

Fashion-MNIST (Xiao et al., 2017); Kingma & Welling (2013), Auto-Encoding Variational Bayes; Radford et al. (2015), DCGAN; Keras Fashion-MNIST documentation (keras.io/api/datasets/fashion_mnist/); TensorFlow API (tensorflow.org/api_docs/python/tf/keras); Hugging Face Diffusers (huggingface.co/docs/diffusers); Stable Diffusion v1.5 model card (huggingface.co/stable-diffusion-v1-5/stable-diffusion-v1-5).